

Insurance

Homework #4

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1 Introduction

We model two outcomes in a auto insurance outcome variables: (1) whether the person was in a car crash (binary) and (2) the amount of money it will cost if the person does crash their car (continuous). Our objective is to build multiple linear regression and binary logistic regression models on the training data to predict the probability that a person will crash their car and also predict the cost of the crash if it occurs.

2 Data Exploration and Cleaning

```
# Load the training and evaluation datasets
insurance_train <- read_csv("./data/insurance_training_data.csv")
insurance_eval <- read_csv("./data/insurance-evaluation-data.csv")

# Count rows and columns
nrows <- insurance_train |> nrow()
ncols <- insurance_train |> length()

# Get missingness
perc_complete <- round(insurance_train |> n_complete() / nrows / ncols * 100, 2)
```

The insurance training data set contains 8161 rows and 26 features, and is 98.87 percent complete. We will need to address the missingness in the data preparation section.

There are several variables in the dataset coded as characters. There are several currency variables (INCOME, HOME_VAL, BLUEBOOK, OLDCLAIM) that should be changed to numeric variables. The rest

of the character variables should be encoded as dummy variables for modeling in the following chunk. We also rename some variables for clarity. We also remove the `index` variable as it is not a predictor and will not be used in modeling.

```
# Convert currency variables to numeric by removing "$" and converting to numeric
currency_vars <- c("INCOME", "HOME_VAL", "BLUEBOOK", "OLDCLAIM")

# Character columns to one-hot encode (exclude currency columns)
char_vars <- insurance_train |>
  select(where(is.character)) |>
  names() |>
  setdiff(currency_vars)

insurance_train_dummies <- insurance_train |>
  mutate(
    # Convert currency variables to numeric
    across(all_of(currency_vars), parse_number)
  ) |>
  fastDummies::dummy_cols(
    select_columns = char_vars,
    remove_selected_columns = TRUE,
    remove_first_dummy = TRUE
  ) |>
  clean_names() |>
  rename(
    female = sex_z_f,
    education_phd = education_ph_d,
    education_high_school = education_z_high_school,
    car_type_suv = car_type_z_suv,
    job_blue_collar = job_z_blue_collar,
    urbanicity_highly_rural = urbanicity_z_highly_rural_rural
  ) |>
  select(-index)

insurance_eval_dummies <- insurance_eval |>
  mutate(
    across(all_of(currency_vars), parse_number)
  ) |>
  fastDummies::dummy_cols(
```

```

    select_columns = char_vars,
    remove_selected_columns = TRUE,
    remove_first_dummy = TRUE
  ) |>
  clean_names() |>
  rename(
    female = sex_z_f,
    education_phd = education_ph_d,
    education_high_school = education_z_high_school,
    car_type_suv = car_type_z_suv,
    job_blue_collar = job_z_blue_collar,
    urbanicity_highly_rural = urbanicity_z_highly_rural_rural
  ) |>
  select(-index)

```

In previewing summary statistics for all variables, some of the minimum values appear to be in error. For example, `car_age` has a negative value, and there are a high number of 0 values in `car_age`, `home_val`, which may be data entry errors—we will replace them missing values.

```

insurance_train_dummies |> summary()

insurance_train_dummies <- insurance_train_dummies |>
  mutate(
    car_age = ifelse(car_age <= 0, NA, car_age),
    home_val = ifelse(home_val == 0, NA, home_val)
  )

```

Table 1: Summary statistics for all variables in the training dataset

Variable	Completeness						
	rate	Mean	Min	P25	P50	P75	Max
target_flag	1.00	0.260	0	0	0	1	1
target_amt	1.00	1500.000	0	0	0	1000	110000
kidsdriv	1.00	0.170	0	0	0	0	4
age	1.00	45.000	16	39	45	51	81
homekids	1.00	0.720	0	0	0	1	5
voj	0.94	10.000	0	9	11	13	23
income	0.95	62000.000	0	28000	54000	86000	370000
home_val	0.66	220000.000	50000	150000	210000	270000	890000

Table 1: Summary statistics for all variables in the training dataset

Variable	Completeness						
	rate	Mean	Min	P25	P50	P75	Max
travtime	1.00	33.000	5	22	33	44	140
bluebook	1.00	16000.000	1500	9300	14000	21000	70000
tif	1.00	5.400	1	1	4	7	25
oldclaim	1.00	4000.000	0	0	0	4600	57000
clm_freq	1.00	0.800	0	0	0	2	5
mvr_pts	1.00	1.700	0	0	1	3	13
car_age	0.94	8.300	1	1	8	12	28
parent1_yes	1.00	0.130	0	0	0	0	1
mstatus_z_no	1.00	0.400	0	0	0	1	1
female	1.00	0.540	0	0	1	1	1
education_bachelors	1.00	0.270	0	0	0	1	1
education_masters	1.00	0.200	0	0	0	0	1
education_phd	1.00	0.089	0	0	0	0	1
educa-	1.00	0.290	0	0	0	1	1
tion_high_school							
job_doctor	0.94	0.032	0	0	0	0	1
job_home_maker	0.94	0.084	0	0	0	0	1
job_lawyer	0.94	0.110	0	0	0	0	1
job_manager	0.94	0.130	0	0	0	0	1
job_professional	0.94	0.150	0	0	0	0	1
job_student	0.94	0.093	0	0	0	0	1
job_blue-collar	0.94	0.240	0	0	0	0	1
job_na	1.00	0.064	0	0	0	0	1
car_use_private	1.00	0.630	0	0	1	1	1
car_type_panel_truck	1.00	0.083	0	0	0	0	1
car_type_pickup	1.00	0.170	0	0	0	0	1
car_type_sports_car	1.00	0.110	0	0	0	0	1
car_type_van	1.00	0.092	0	0	0	0	1
car_type_suv	1.00	0.280	0	0	0	1	1
red_car_yes	1.00	0.290	0	0	0	1	1
revoked_yes	1.00	0.120	0	0	0	0	1
urbanic-	1.00	0.200	0	0	0	0	1
ity_highly_rural							

Table 1 presents descriptive characteristics of the training data. The target variables are

`target_flag`, which represents whether the person was in a car crash (1) or not (0), and `target_amt`, which represents the cost of the crash if it occurs. The other variables represent various information on auto insurance for different individuals.

```
# Get correlation matrix
insurance_train_cor <- insurance_train_dummies |>
  select(where(is.numeric)) |>
  cor(use = "pairwise.complete.obs")

insurance_train_cor_plot <- insurance_train_cor
insurance_train_cor_plot[abs(insurance_train_cor_plot) < 0.2] <- NA

corrplot(insurance_train_cor_plot, method = "circle", type = "upper", sig.level = 0.05, insig = 0.05)

most_corr_w_target_flag <- insurance_train_cor |>
  as.data.frame() |>
  rownames_to_column(var = "Variable") |>
  filter(Variable != "target_flag", Variable != "target_amt") |>
  transmute(
    Variable,
    Correlation = target_flag,
    Abs_Correlation = abs(target_flag)
  ) |>
  arrange(desc(Abs_Correlation)) |>
  filter(Abs_Correlation > 0.2) |> pull(Variable)

most_corr_w_target_amt <- insurance_train_cor |>
  as.data.frame() |>
  rownames_to_column(var = "Variable") |>
  filter(Variable != "target_flag", Variable != "target_amt") |>
  transmute(
    Variable,
    Correlation = target_amt,
    Abs_Correlation = abs(target_amt)
  ) |>
  arrange(desc(Abs_Correlation)) |>
  filter(Abs_Correlation > 0.2) |>
  pull(Variable)
```

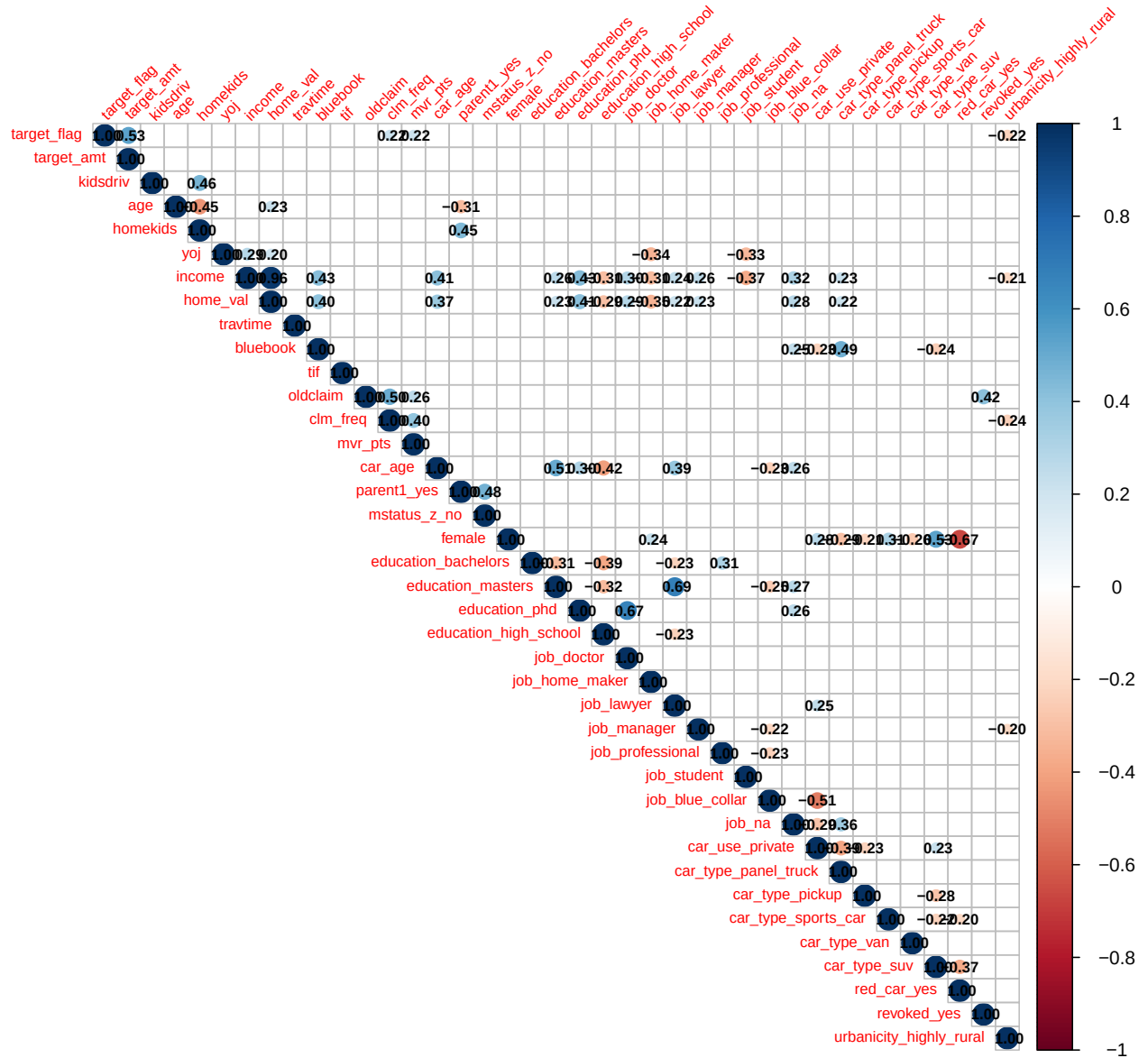


Figure 1: Correlation matrix of all variables in the training dataset. Note: Correlations with absolute value less than 0.2 are suppressed.

The correlation between all variables are shown Figure 1. The variables most correlated with `target_flag`, whether there was a car crash, (besides the amount the crash costed) are `urbanicity_highly_rural`, `mvr_pts`, `clm_freq`. This means that people who work or live in a more rural area are less likely to be in a crash, and people who have more motor vehicle record points or have more claims in the past 5 years are more likely to be in a crash. There were no predictor variables strongly correlated with `target_amt`, the cost if the crash occurs.

Lastly, for continuous variables, we also looked at the distribution of each variable to check for skewness and outliers.

```
# Keep only continuous
continuous_vars <- insurance_train_dummies |>
  select(where(is.numeric)) |>
  select(where(~ dplyr::n_distinct(stats::na.omit(.x)) > 10)) |>
  names()

insurance_train_dummies |>
  select(all_of(continuous_vars)) |>
  pivot_longer(cols = everything(), names_to = "variable", values_to = "value") |>
  ggplot(aes(x = value)) +
  geom_histogram(bins = 30, fill = "steelblue", color = "white") +
  facet_wrap(~ variable, scales = "free", ncol = 4) +
  theme_minimal() +
  theme(panel.grid = element_blank()) +
  labs(x = NULL, y = "Count")
```

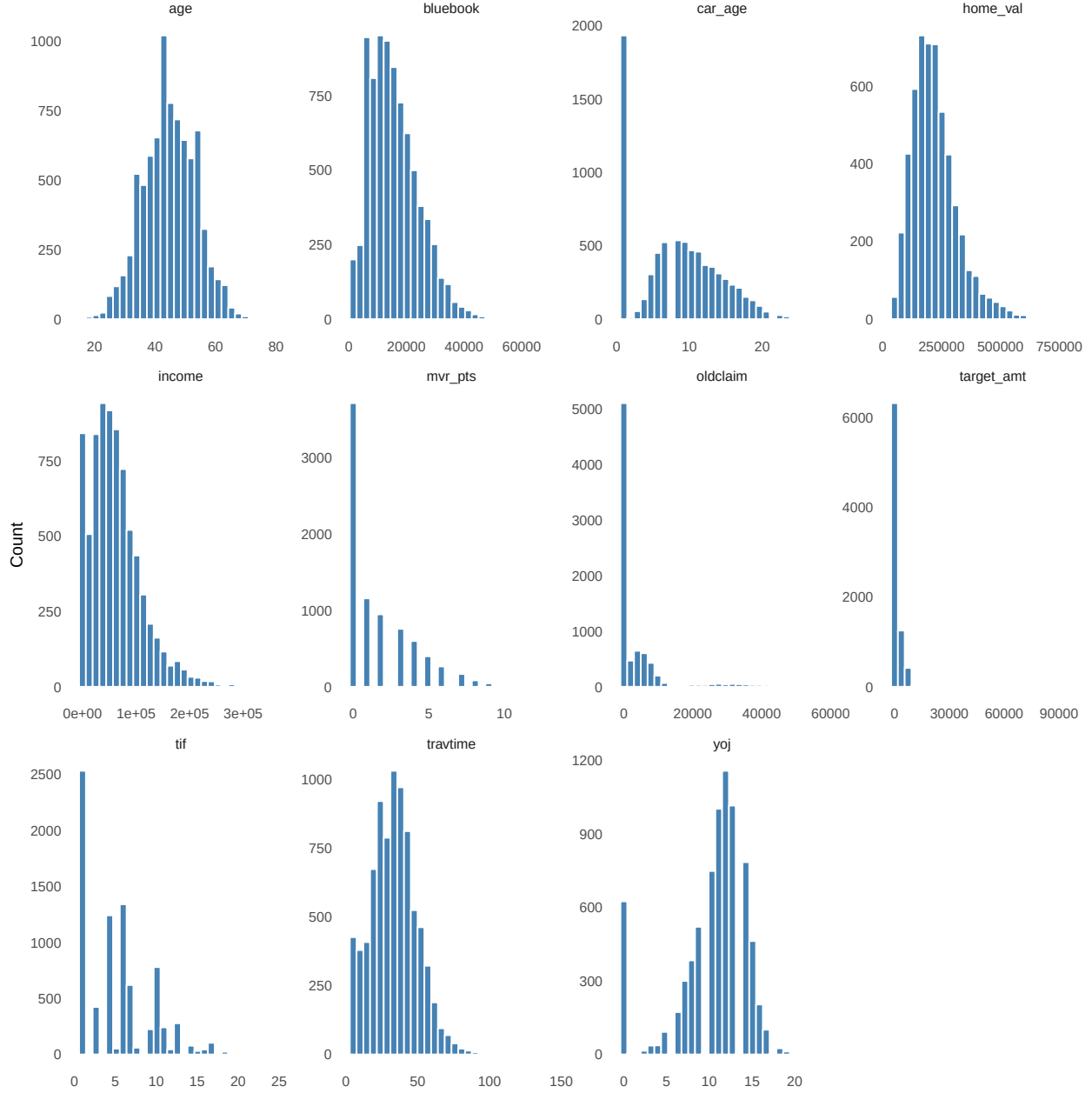


Figure 2: Distribution of continuous variables in the training dataset.

There are many 1-year old cars in the dataset, showing a high spike at 1 in the **car_age** variable. Dollar amounts such as **bluebook**, **home_val**, **income**, and **target_amt** are right-skewed, which is expected for financial variables. The same goes for travel time (**travtime**). Most people have no motor vehicle record points (**mvr_pts**) or claims in the past 5 years (**oldclaim**), which is expected for a general population. Years on the job (**yoj**) is almost normally distributed, but has a spike at 0, which may represent people who are unemployed or have been on the job for less than a year.

3 Data Preparation

The goal of this section is to prepare the insurance training and evaluation data for modeling. The assignment requires transformations of the original variables or the creation of new variables when useful. In this dataset, preparation included missing value treatment, missingness flags, mathematical transformations, and derived variables.

```
# Identify numeric predictor variables
numeric_vars <- insurance_train_dummies |>
  select(where(is.numeric)) |>
  select(-target_flag, -target_amt) |>
  names()

# Create missingness flags before imputation
insurance_train_prepared <- insurance_train_dummies |>
  mutate(
    age_missing = ifelse(is.na(age), 1, 0),
    car_age_missing = ifelse(is.na(car_age), 1, 0),
    home_val_missing = ifelse(is.na(home_val), 1, 0),
    income_missing = ifelse(is.na(income), 1, 0),
    yoj_missing = ifelse(is.na(yoj), 1, 0)
  )

insurance_eval_prepared <- insurance_eval_dummies |>
  mutate(
    age_missing = ifelse(is.na(age), 1, 0),
    car_age_missing = ifelse(is.na(car_age), 1, 0),
    home_val_missing = ifelse(is.na(home_val), 1, 0),
    income_missing = ifelse(is.na(income), 1, 0),
    yoj_missing = ifelse(is.na(yoj), 1, 0)
  )

# Median imputation using training medians
for (v in numeric_vars) {
  med_val <- median(insurance_train_prepared[[v]], na.rm = TRUE)

  insurance_train_prepared[[v]][is.na(insurance_train_prepared[[v]])] <- med_val
  insurance_eval_prepared[[v]][is.na(insurance_eval_prepared[[v]])] <- med_val
}
```

```
# Create transformed and engineered variables
insurance_train_prepared <- insurance_train_prepared |>
  mutate(
    log_income = log1p(income),
    log_home_val = log1p(home_val),
    log_bluebook = log1p(bluebook),
    log_oldclaim = log1p(oldclaim),
    log_travtime = log1p(travtime),

    claim_rate = oldclaim / (clm_freq + 1),
    car_value_age_ratio = bluebook / (car_age + 1),
    risk_points_claims = mvr_pts + clm_freq,

    age_bucket = case_when(
      age < 25 ~ "young_driver",
      age >= 25 & age < 60 ~ "adult_driver",
      age >= 60 ~ "older_driver",
      TRUE ~ "unknown"
    )
  )

insurance_eval_prepared <- insurance_eval_prepared |>
  mutate(
    log_income = log1p(income),
    log_home_val = log1p(home_val),
    log_bluebook = log1p(bluebook),
    log_oldclaim = log1p(oldclaim),
    log_travtime = log1p(travtime),

    claim_rate = oldclaim / (clm_freq + 1),
    car_value_age_ratio = bluebook / (car_age + 1),
    risk_points_claims = mvr_pts + clm_freq,

    age_bucket = case_when(
      age < 25 ~ "young_driver",
      age >= 25 & age < 60 ~ "adult_driver",
      age >= 60 ~ "older_driver",
      TRUE ~ "unknown"
    )
  )
```

```
)

# Convert age bucket into dummy variables
insurance_train_prepared <- fastDummies::dummy_cols(
  insurance_train_prepared,
  select_columns = "age_bucket",
  remove_selected_columns = TRUE,
  remove_first_dummy = TRUE
)

insurance_eval_prepared <- fastDummies::dummy_cols(
  insurance_eval_prepared,
  select_columns = "age_bucket",
  remove_selected_columns = TRUE,
  remove_first_dummy = TRUE
)

# Confirm no missing values remain
sum(is.na(insurance_train_prepared))
```

```
[1] 0
```

```
sum(is.na(insurance_eval_prepared))
```

```
[1] 4282
```

3.1 Data Splits

First, we will split our dataset into a training and testing set. We will use the training set to build our models and the testing set to evaluate their performance. The training set will be used to fit the models, while the testing set will be used to assess how well the models generalize to new data.

```
set.seed(19940211)
train_indices <- sample(nrow(insurance_train_prepared), size = 0.8 * nrow(insurance_train_prepared))
insurance_train_split <- insurance_train_prepared[train_indices, ]
insurance_test_split <- insurance_train_prepared[-train_indices, ]
```

4 Check assumptions for Binary logistic regression

1. Binary Outcome

```
# Check assumptions for binary logistic regression

#| label: assumption-binary-outcome
#| echo: true

# Confirming that the target is binary
insurance_train_split |>
  count(target_flag) |>
  mutate(prop = n / sum(n))
```

```
# A tibble: 2 x 3
  target_flag     n prop
  <dbl> <int> <dbl>
1         0  4807 0.736
2         1  1721 0.264
```

The response variable `target_flag` is confirmed to be binary, taking two values (0: no car crash) and (1: car crash occurred). In the training split, approximately 26% of policyholders were involved in a crash and 74% were not. Unlike the crime dataset in Homework 3, there is a moderate class imbalance here, which means accuracy alone may be a misleading performance metric. We will rely on sensitivity, specificity, and AUC-ROC when evaluating our models.

2. Independence of Observations

Each record in the dataset represents a distinct policyholder, and there is no repeated-measures or clustered structure in the data. Therefore, the independence of observations assumption is satisfied by design and requires no further adjustment.

3. No Multicollinearity (VIF Check)

```
library(car)

# Fit a reference logistic model with core untransformed predictors
model_ref <- glm(
  target_flag ~ age + homekids + yoj + income + home_val + travtime +
```

```

      bluebook + tif + oldclaim + clm_freq + mvr_pts + car_age,
    data = insurance_train_split,
    family = binomial
  )

# Compute VIF
vif_vals <- vif(model_ref) |>
  as.data.frame() |>
  rownames_to_column("Variable") |>
  rename(VIF = 2) |>
  arrange(desc(VIF))

vif_vals |> knitr::kable(caption = "Variance Inflation Factors for Reference Model")

```

Table 2: Variance Inflation Factors for Reference Model

Variable	VIF
income	2.452752
home_val	1.989529
clm_freq	1.391524
age	1.324525
homekids	1.313457
oldclaim	1.271882
car_age	1.191627
bluebook	1.176350
yoj	1.175710
mvr_pts	1.160473
tif	1.003529
travtime	1.003503

VIF values were calculated for all predictors to check for multicollinearity. A VIF above 10 is considered to be severe, while between 5 and 10 should warrant caution. None of the variables exceed the threshold of 10, indicating multicollinearity is not present. The highest VIF belongs to income (VIF = 2.45), followed by home_val (VIF = 1.99). All remaining predictors fall well below 5 and are at acceptable levels of multicollinearity. Therefore, no variables need to be removed on the basis of multicollinearity, and all predictors may be considered for model building.

4. Linearity of Log-Odds (Box-Tidwell Test)

```

# Box-Tidwell test: add interaction of each continuous predictor with its log
# to test whether the log-odds relationship is linear
continuous_vars <- c("age", "homekids", "yoj", "travtime", "bluebook",
                    "tif", "oldclaim", "clm_freq", "mvr_pts", "car_age")

# Create interaction terms x * log(x) for each continuous predictor
# (use log1p to handle zeros)
insurance_bt <- insurance_train_split |>
  mutate(across(all_of(continuous_vars), ~ . * log1p(.), .names = "{.col}_logx"))

bt_formula <- as.formula(paste(
  "target_flag ~",
  paste(continuous_vars, collapse = " + "),
  "+",
  paste(paste0(continuous_vars, "_logx"), collapse = " + ")
))

model_bt <- glm(bt_formula, data = insurance_bt, family = binomial)

# Extract p-values for the interaction terms only
bt_results <- summary(model_bt)$coefficients |>
  as.data.frame() |>
  rownames_to_column("Term") |>
  filter(str_detect(Term, "_logx")) |>
  mutate(Variable = str_remove(Term, "_logx"),
         Significant = `Pr(>|z|)` < 0.05) |>
  select(Variable, Estimate, `Pr(>|z|)`, Significant) |>
  mutate(across(where(is.numeric), ~ signif(., 3)))

bt_results |> knitr::kable(caption = "Box-Tidwell Test for Linearity of Log-Odds")

```

Table 3: Box-Tidwell Test for Linearity of Log-Odds

Variable	Estimate	Pr(> z)	Significant
age	1.22e-01	3.00e-07	TRUE
homekids	-2.62e-01	3.05e-03	TRUE
yoj	-1.05e-02	5.80e-01	FALSE
travtime	-1.73e-02	2.14e-03	TRUE
bluebook	3.78e-05	1.98e-04	TRUE

Variable	Estimate	Pr(> z)	Significant
tif	3.85e-02	4.80e-02	TRUE
oldclaim	-1.02e-05	3.02e-01	FALSE
clm_freq	-4.91e-01	1.35e-05	TRUE
mvr_pts	1.40e-02	6.55e-01	FALSE
car_age	1.03e-02	4.74e-01	FALSE

The Box-Tidwell test was used to assess whether each continuous predictor has a linear relationship with the log-odds of a car crash. A significant p-value (< 0.05) for the interaction term indicates a violation of this assumption. Six variables were flagged as significant: age ($p < 0.001$), homekids ($p = 0.003$), travtime ($p = 0.002$), bluebook ($p < 0.001$), tif ($p = 0.048$), and clm_freq ($p < 0.001$). This indicates that these predictors do not have a strictly linear relationship with the log-odds of a car crash. Log-transformed versions of travtime and bluebook (log_travtime, log_bluebook) were already created in the data preparation step and will be used in model building. For age, homekids, tif, and clm_freq, we will consider log1p transformations or binning strategies during model building to address the non-linearity. The remaining four predictors — yoj, oldclaim, mvr_pts, and car_age — did not show evidence of violating the linearity of log-odds assumption and may enter models in their original form.

5 Check Assumptions for Linear Regression

For the linear regression model, we model `target_amt` only on records where a crash occurred (`target_flag == 1`), since a value of zero indicates no crash rather than an actual claim cost of zero.

```
# Subset to only crashed records for linear regression assumption checks
insurance_train_crash <- insurance_train_split |>
  filter(target_flag == 1)

nrow(insurance_train_crash)
```

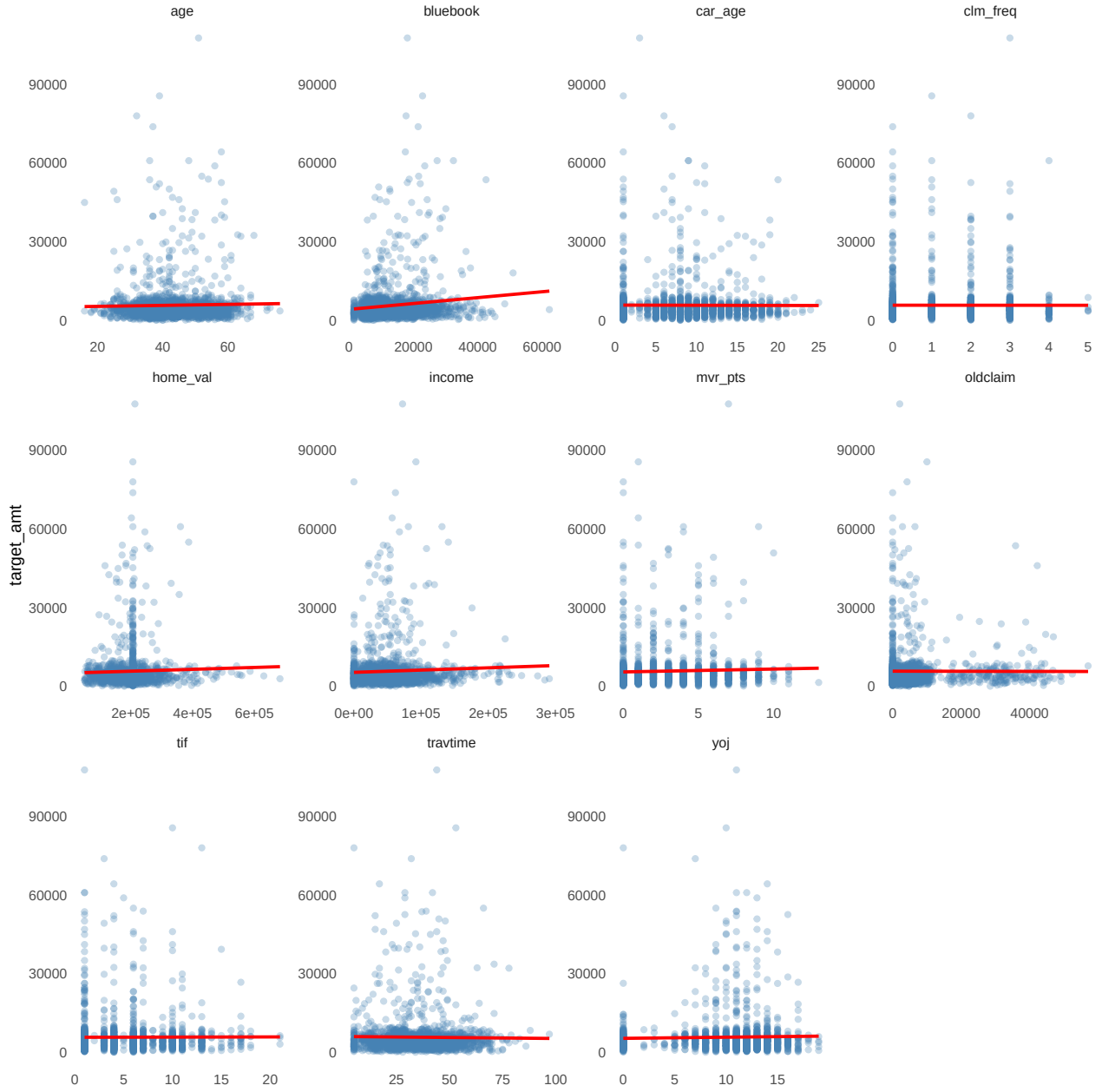
```
[1] 1721
```

1. Linearity

```
continuous_vars_lm <- c("age", "yoj", "income", "home_val", "travtime",
  "bluebook", "tif", "oldclaim", "clm_freq",
```

```
      "mvr_pts", "car_age")

insurance_train_crash |>
  select(all_of(continuous_vars_lm), target_amt) |>
  pivot_longer(-target_amt, names_to = "variable", values_to = "value") |>
  ggplot(aes(x = value, y = target_amt)) +
  geom_point(alpha = 0.3, color = "steelblue") +
  geom_smooth(method = "lm", se = FALSE, color = "red") +
  facet_wrap(~ variable, scales = "free", ncol = 4) +
  theme_minimal() +
  theme(panel.grid = element_blank()) +
  labs(x = NULL, y = "target_amt")
```


Figure 3: Scatterplots of continuous predictors vs. `target_amt`

The scatterplots show the relationship between each continuous predictor and `target_amt`. Most predictors show a weak or near-flat linear relationship with the response, which is consistent with the earlier finding that no predictor variables were strongly correlated with `target_amt` in the correlation matrix. The only exception is `bluebook`, which shows a slight positive trend, suggesting that more expensive vehicles are associated with higher crash costs. The heavy concentration of points at lower values of `target_amt` across all predictors reflects the right-skewed distribution of the response variable noted during data exploration. Log transformations of `bluebook`, `income`, `oldclaim`, and `travtime` created in the data preparation step will be used during model building.

to address this.

2. Independence of Observations

Each record in the dataset represents a distinct policyholder involved in a crash. There is no repeated-measures or clustered structure in the data. Therefore, the independence of observations assumption is satisfied by design and requires no further adjustment.

3. Homoscedasticity (Constant Variance)

```
# Fit a reference linear model on crash records only
model_ref_lm <- lm(
  target_amt ~ age + yoj + income + home_val + travtime +
    bluebook + tif + oldclaim + clm_freq + mvr_pts + car_age,
  data = insurance_train_crash
)

# Residuals vs fitted plot
plot(model_ref_lm, which = 1,
     main = "Residuals vs Fitted",
     pch = 20, col = "steelblue")
```

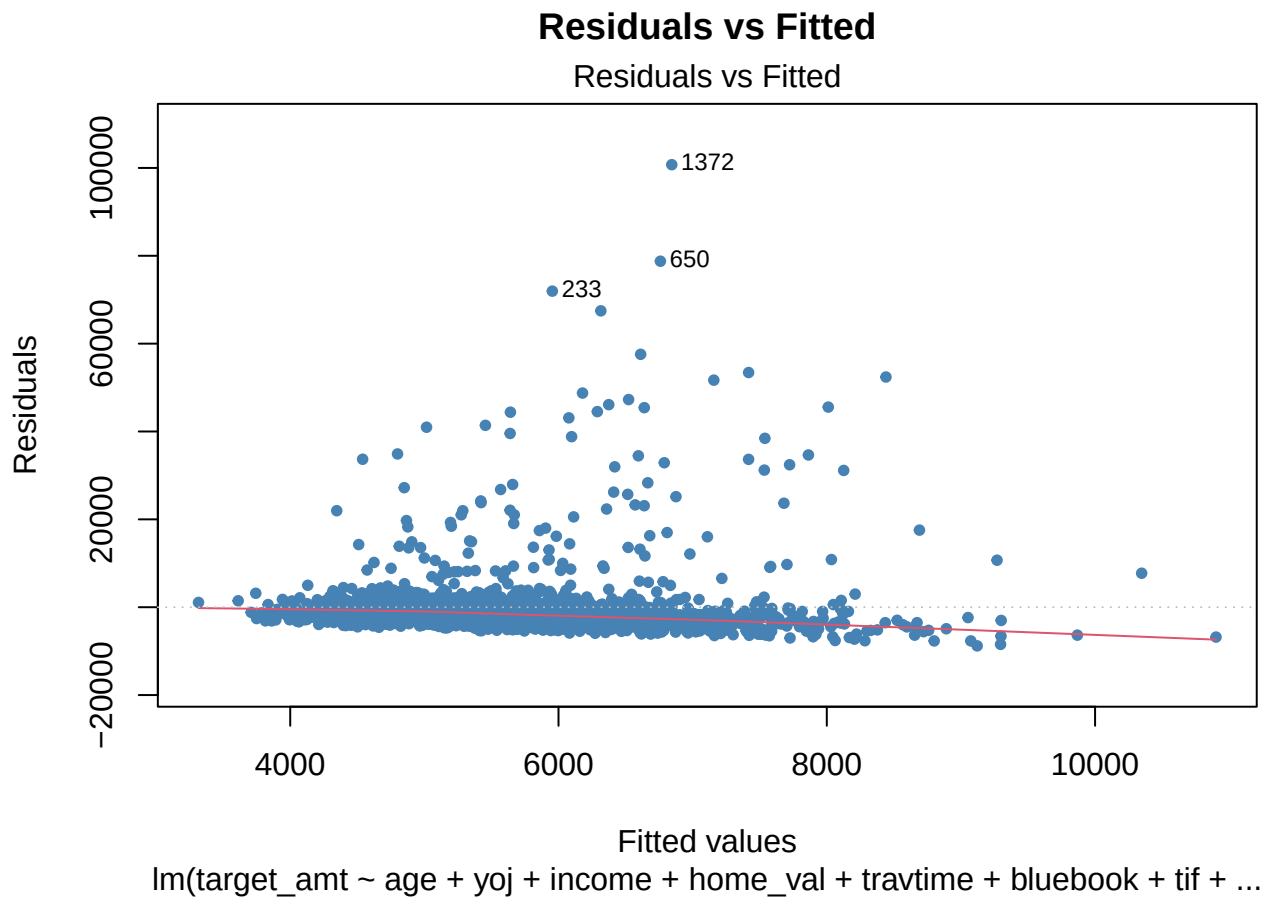


Figure 4: Residuals vs. Fitted values plot for homoscedasticity check

The residuals vs. fitted plot shows a clear fanning pattern, where the spread of residuals increases as fitted values increase. This indicates heteroscedasticity — the variance of residuals is not constant across fitted values. Several high-leverage outliers (observations 233, 650, and 1372) are visible with residuals exceeding \$60,000, pulling the fitted line downward at higher values. A log transformation of `target_amt` will be considered during model building to stabilize the variance.

4. Normality of Residuals

```
# Q-Q plot
plot(model_ref_lm, which = 2,
     main = "Normal Q-Q Plot of Residuals",
     pch = 20, col = "steelblue")
```

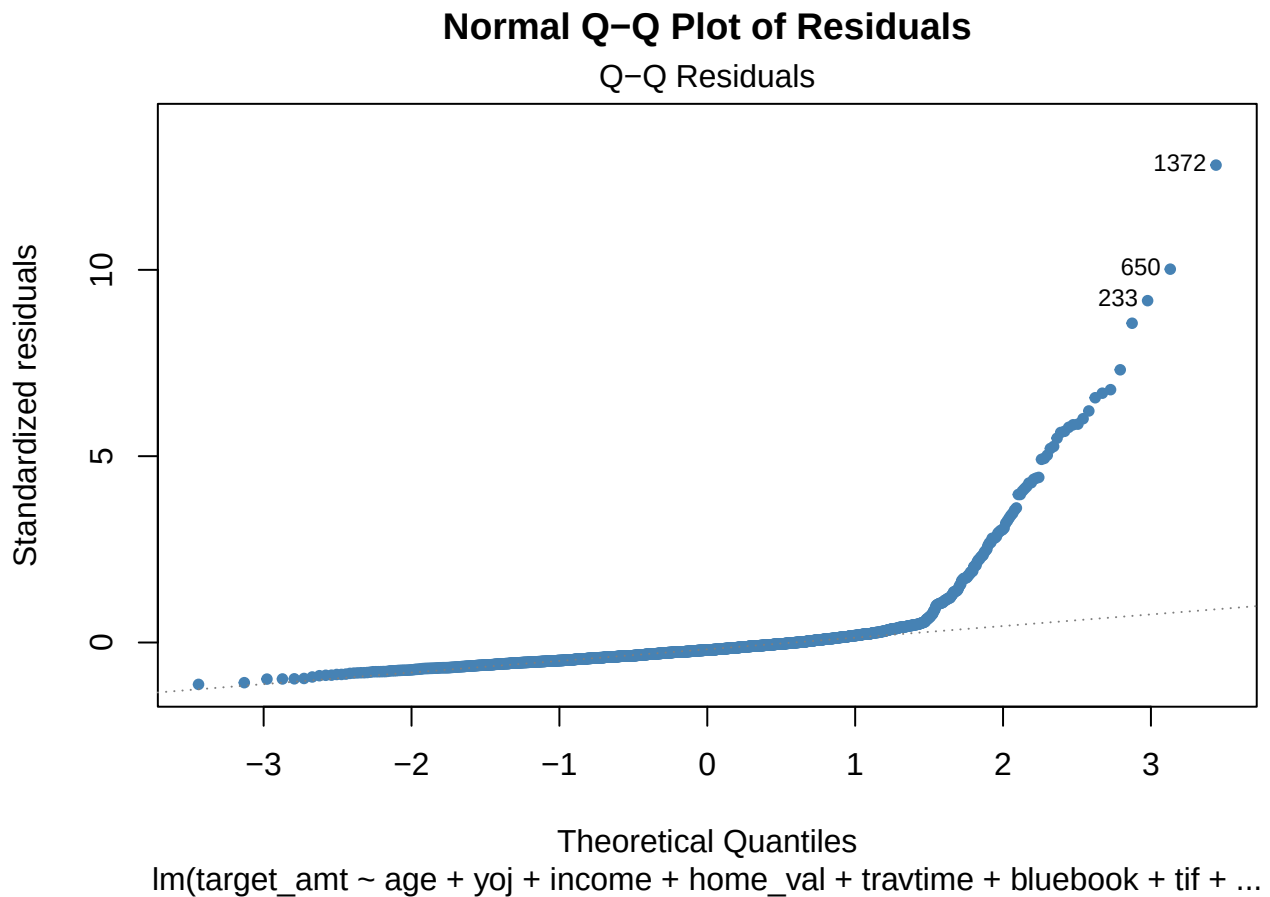


Figure 5: Q-Q plot of residuals for normality check

```
# Shapiro-Wilk test on a sample (max 5000 observations)
set.seed(19940211)
residuals_sample <- residuals(model_ref_lm)

if (length(residuals_sample) > 5000) {
  residuals_sample <- sample(residuals_sample, 5000)
}

shapiro.test(residuals_sample)
```

Shapiro-Wilk normality test

data: residuals_sample

W = 0.47904, p-value < 2.2e-16

The Q-Q plot and Shapiro-Wilk test are used to assess whether the residuals are normally dis-

tributed. Departures from the diagonal line in the Q-Q plot indicate non-normality. The Shapiro-Wilk test confirms a violation of this assumption ($W = 0.479$, $p < 0.001$), indicating that the residuals are significantly non-normal. The same three outliers identified in the residuals vs. fitted plot (observations 233, 650, and 1372) are again visible with standardized residuals exceeding 8. A log transformation of `target_amt` will be applied during model building to address both the non-normality and heteroscedasticity identified in assumptions 3 and 4.

6 Build models

We will build models for two separate outcomes:

1. **target_flag** (binary): Whether the person was in a car crash (1) or not (0)
2. **target_amt** (continuous): The amount of money the crash cost if it occurs

For each outcome, we compare three models as per the textbook methods: - Model 1: Full model with all predictors - Model 2: Reduced model (removing highly correlated or insignificant variables) - Model 3: Backward stepwise selection model

6.1 Model 1. Full Logistic and Linear Models

As shown below, the most significant predictors for `target_flag` (whether a crash occurred) include variables related to the driver's profile and history. The AIC for this logistic model is `r\AIC(model1_logistic)`.

Similarly, for `target_amt` (cost of crash), the full linear model identifies significant predictors. The R-squared for this model is `r\ summary(model1_linear)$r.squared`.

```
# Model 1: Full Logistic Model (target_flag)
# This model uses all available numeric predictors
model1_logistic <- glm(
  formula = target_flag ~ .,
  data = insurance_train_split |> select(-target_amt),
  family = binomial
)

cat("\n=== Full Logistic Model Summary ===\n")
```

```
=== Full Logistic Model Summary ===
```

```
summary(model1_logistic)
```

Call:

```
glm(formula = target_flag ~ ., family = binomial, data = select(insurance_train_split,
  -target_amt))
```

Coefficients: (1 not defined because of singularities)

	Estimate	Std. Error	z value	Pr(> z)	
(Intercept)	7.944e+00	4.389e+00	1.810	0.070287	.
kidsdriv	4.407e-01	6.883e-02	6.403	1.52e-10	***
age	-7.728e-03	5.236e-03	-1.476	0.139948	
homekids	6.359e-03	4.222e-02	0.151	0.880264	
yoj	1.437e-02	1.258e-02	1.142	0.253363	
income	-1.855e-06	1.521e-06	-1.220	0.222471	
home_val	1.026e-06	1.940e-06	0.529	0.597085	
travtime	9.614e-03	6.062e-03	1.586	0.112719	
bluebook	5.090e-06	1.249e-05	0.407	0.683678	
tif	-5.336e-02	8.322e-03	-6.411	1.44e-10	***
oldclaim	-2.687e-05	1.382e-05	-1.945	0.051786	.
clm_freq	1.617e-02	6.380e-02	0.253	0.799915	
mvr_pts	9.445e-02	1.587e-02	5.952	2.65e-09	***
car_age	4.851e-03	1.200e-02	0.404	0.685993	
parent1_yes	4.088e-01	1.238e-01	3.301	0.000962	***
mstatus_z_no	4.909e-01	9.584e-02	5.122	3.02e-07	***
female	-1.415e-01	1.278e-01	-1.108	0.268011	
education_bachelors	-3.574e-01	1.326e-01	-2.694	0.007049	**
education_masters	-3.316e-01	2.080e-01	-1.594	0.110934	
education_phd	-4.985e-01	2.513e-01	-1.984	0.047281	*
education_high_school	5.154e-02	1.091e-01	0.473	0.636525	
job_doctor	-7.560e-01	3.343e-01	-2.261	0.023736	*
job_home_maker	-5.309e-01	1.844e-01	-2.879	0.003991	**
job_lawyer	-3.305e-01	2.103e-01	-1.572	0.116049	
job_manager	-9.606e-01	1.614e-01	-5.951	2.67e-09	***
job_professional	-2.989e-01	1.404e-01	-2.129	0.033285	*
job_student	-5.214e-01	1.722e-01	-3.028	0.002464	**
job_blue_collar	-1.258e-01	1.213e-01	-1.037	0.299751	
job_na	-3.484e-01	2.242e-01	-1.554	0.120120	
car_use_private	-7.139e-01	1.041e-01	-6.855	7.12e-12	***

car_type_panel_truck	5.357e-01	1.873e-01	2.860	0.004237	**
car_type_pickup	5.797e-01	1.139e-01	5.091	3.57e-07	***
car_type_sports_car	9.700e-01	1.497e-01	6.482	9.07e-11	***
car_type_van	6.665e-01	1.439e-01	4.633	3.60e-06	***
car_type_suv	7.805e-01	1.276e-01	6.118	9.46e-10	***
red_car_yes	-1.006e-01	9.876e-02	-1.019	0.308307	
revoked_yes	9.370e-01	1.048e-01	8.941	< 2e-16	***
urbanicity_highly_rural	-2.353e+00	1.278e-01	-18.416	< 2e-16	***
age_missing	1.978e+00	1.230e+00	1.608	0.107754	
car_age_missing	1.679e-01	1.368e-01	1.227	0.219674	
home_val_missing	3.324e-01	9.554e-02	3.480	0.000502	***
income_missing	1.810e-02	1.500e-01	0.121	0.903968	
yoj_missing	6.266e-02	1.439e-01	0.435	0.663252	
log_income	-7.355e-02	2.267e-02	-3.244	0.001177	**
log_home_val	-4.556e-01	3.831e-01	-1.189	0.234347	
log_bluebook	-4.049e-01	1.392e-01	-2.908	0.003633	**
log_oldclaim	8.193e-02	1.980e-02	4.138	3.51e-05	***
log_travtime	2.118e-01	1.727e-01	1.226	0.220103	
claim_rate	-3.106e-06	3.730e-05	-0.083	0.933638	
car_value_age_ratio	1.851e-05	1.850e-05	1.000	0.317098	
risk_points_claims	NA	NA	NA	NA	
age_bucket_older_driver	7.900e-01	1.886e-01	4.190	2.79e-05	***
age_bucket_young_driver	4.971e-01	3.333e-01	1.492	0.135820	

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 7531.0 on 6527 degrees of freedom
 Residual deviance: 5748.1 on 6476 degrees of freedom
 AIC: 5852.1

Number of Fisher Scoring iterations: 5

```
# Model 1: Full Linear Model (target_amt)
# This model uses all available numeric predictors
model1_linear <- lm(
  formula = target_amt ~ .,
  data = insurance_train_split |> select(-target_flag)
```

```
)

cat("\n=== Full Linear Model Summary ===\n")
```

```
=== Full Linear Model Summary ===
```

```
summary(model1_linear)
```

Call:

```
lm(formula = target_amt ~ ., data = select(insurance_train_split,
      -target_flag))
```

Residuals:

Min	1Q	Median	3Q	Max
-5476	-1694	-760	387	103433

Coefficients: (1 not defined because of singularities)

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	9.078e+02	7.454e+03	0.122	0.90307
kidsdriv	3.108e+02	1.287e+02	2.414	0.01579 *
age	-1.282e-01	9.262e+00	-0.014	0.98896
homekids	3.280e+01	7.557e+01	0.434	0.66433
yoj	-1.116e+00	2.189e+01	-0.051	0.95936
income	-2.254e-03	2.672e-03	-0.843	0.39899
home_val	-6.342e-04	3.161e-03	-0.201	0.84098
travtime	4.353e+00	1.018e+01	0.427	0.66908
bluebook	9.426e-04	2.091e-02	0.045	0.96405
tif	-4.269e+01	1.400e+01	-3.049	0.00230 **
oldclaim	-6.465e-02	2.699e-02	-2.396	0.01662 *
clm_freq	1.338e+02	1.288e+02	1.038	0.29910
mvr_pts	1.633e+02	3.062e+01	5.333	9.97e-08 ***
car_age	-1.753e+01	2.066e+01	-0.848	0.39626
parent1_yes	5.603e+02	2.313e+02	2.422	0.01545 *
mstatus_z_no	5.819e+02	1.679e+02	3.467	0.00053 ***
female	-3.282e+02	2.135e+02	-1.537	0.12437
education_bachelors	-2.888e+02	2.391e+02	-1.208	0.22723
education_masters	3.296e+00	3.498e+02	0.009	0.99248

education_phd	-2.056e+02	4.151e+02	-0.495	0.62046
education_high_school	-1.813e+02	1.997e+02	-0.908	0.36405
job_doctor	-9.151e+02	5.033e+02	-1.818	0.06908 .
job_home_maker	-4.712e+02	3.171e+02	-1.486	0.13727
job_lawyer	-4.518e+02	3.503e+02	-1.290	0.19716
job_manager	-1.158e+03	2.726e+02	-4.247	2.20e-05 ***
job_professional	-1.423e+02	2.485e+02	-0.573	0.56696
job_student	-4.002e+02	3.120e+02	-1.283	0.19962
job_blue-collar	-8.843e+01	2.221e+02	-0.398	0.69060
job_na	-7.612e+02	3.922e+02	-1.941	0.05235 .
car_use_private	-8.037e+02	1.889e+02	-4.255	2.12e-05 ***
car_type_panel_truck	2.656e+02	3.251e+02	0.817	0.41401
car_type_pickup	2.608e+02	1.948e+02	1.339	0.18074
car_type_sports_car	1.009e+03	2.553e+02	3.952	7.85e-05 ***
car_type_van	5.800e+02	2.449e+02	2.369	0.01787 *
car_type_suv	6.435e+02	2.084e+02	3.088	0.00202 **
red_car_yes	-8.997e+01	1.724e+02	-0.522	0.60185
revoked_yes	5.471e+02	2.003e+02	2.731	0.00633 **
urbanicity_highly_rural	-1.623e+03	1.617e+02	-10.038	< 2e-16 ***
age_missing	1.162e+03	2.095e+03	0.555	0.57892
car_age_missing	7.637e+01	2.433e+02	0.314	0.75362
home_val_missing	1.516e+02	1.741e+02	0.871	0.38383
income_missing	1.257e+02	2.608e+02	0.482	0.62983
yoj_missing	3.243e+02	2.523e+02	1.285	0.19869
log_income	-3.847e+01	4.027e+01	-0.955	0.33946
log_home_val	-1.993e+01	6.518e+02	-0.031	0.97561
log_bluebook	1.353e+02	2.453e+02	0.551	0.58139
log_oldclaim	2.420e+01	3.927e+01	0.616	0.53773
log_travtime	1.811e+02	2.828e+02	0.640	0.52189
claim_rate	1.335e-01	7.295e-02	1.830	0.06727 .
car_value_age_ratio	1.344e-02	3.263e-02	0.412	0.68031
risk_points_claims	NA	NA	NA	NA
age_bucket_older_driver	5.248e+02	3.444e+02	1.524	0.12758
age_bucket_young_driver	3.244e+02	6.563e+02	0.494	0.62111

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 4643 on 6476 degrees of freedom

Multiple R-squared: 0.0709, Adjusted R-squared: 0.06359

F-statistic: 9.69 on 51 and 6476 DF, p-value: < 2.2e-16

6.2 Model 2. Reduced Models (Addressing Multicollinearity)

In the Data Exploration section, we identified several variables with high correlations. For the reduced models, we remove highly correlated predictors while retaining theoretically important ones.

The AIC for the reduced logistic model is `r\ AIC(model2_logistic)`, and the R-squared for the reduced linear model is `r\ summary(model2_linear)$r.squared`.

```
# Model 2: Reduced Logistic Model
# Keep only important predictors based on correlation analysis
reduced_vars <- c("kidsdriv", "age", "homekids", "yoj", "income", "home_val",
                  "travtime", "bluebook", "tif", "oldclaim", "clm_freq", "mvr_pts", "car_age")

model2_logistic <- glm(
  formula = as.formula(paste("target_flag ~", paste(reduced_vars, collapse = "+"))),
  data = insurance_train_split,
  family = binomial
)

cat("\n=== Reduced Logistic Model Summary ===\n")
```

=== Reduced Logistic Model Summary ===

```
summary(model2_logistic)
```

Call:

```
glm(formula = as.formula(paste("target_flag ~", paste(reduced_vars,
  collapse = "+"))), family = binomial, data = insurance_train_split)
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)	
(Intercept)	-4.658e-01	2.236e-01	-2.083	0.037259	*
kidsdriv	2.977e-01	6.039e-02	4.930	8.23e-07	***
age	-8.546e-03	3.994e-03	-2.140	0.032383	*
homekids	8.324e-02	3.304e-02	2.519	0.011754	*

```

yoj      -1.752e-02  7.679e-03  -2.281  0.022524  *
income   -3.230e-06  1.133e-06  -2.851  0.004353  **
home_val  -7.670e-07  6.134e-07  -1.250  0.211143
travtime   8.276e-03  1.848e-03   4.478  7.52e-06  ***
bluebook  -1.400e-05  4.096e-06  -3.419  0.000629  ***
tif       -4.374e-02  7.562e-03  -5.784  7.28e-09  ***
oldclaim   6.127e-06  3.481e-06   1.760  0.078418  .
clm_freq   2.691e-01  2.868e-02   9.381  < 2e-16  ***
mvr_pts    1.479e-01  1.394e-02  10.616  < 2e-16  ***
car_age    -2.270e-02  6.078e-03  -3.735  0.000188  ***

```

```
---
```

```
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
(Dispersion parameter for binomial family taken to be 1)
```

```

Null deviance: 7531.0  on 6527  degrees of freedom
Residual deviance: 6803.1  on 6514  degrees of freedom
AIC: 6831.1

```

```
Number of Fisher Scoring iterations: 4
```

```

# Model 2: Reduced Linear Model
model2_linear <- lm(
  formula = as.formula(paste("target_amt ~", paste(reduced_vars, collapse = "+"))),
  data = insurance_train_split
)

cat("\n=== Reduced Linear Model Summary ===\n")

```

```
=== Reduced Linear Model Summary ===
```

```
summary(model2_linear)
```

```
Call:
```

```
lm(formula = as.formula(paste("target_amt ~", paste(reduced_vars,
  collapse = "+"))), data = insurance_train_split)
```

Residuals:

Min	1Q	Median	3Q	Max
-4857	-1565	-968	-200	104001

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	1.517e+03	4.385e+02	3.459	0.000546	***
kidsdriv	2.569e+02	1.300e+02	1.977	0.048115	*
age	-4.561e+00	7.936e+00	-0.575	0.565478	
homekids	1.081e+02	6.768e+01	1.598	0.110143	
yoj	-1.700e+01	1.568e+01	-1.084	0.278264	
income	-2.187e-03	2.083e-03	-1.050	0.293755	
home_val	-9.346e-04	1.117e-03	-0.836	0.402940	
travtime	6.382e+00	3.664e+00	1.742	0.081575	.
bluebook	1.626e-02	7.721e-03	2.106	0.035246	*
tif	-3.865e+01	1.419e+01	-2.725	0.006450	**
oldclaim	5.048e-03	7.649e-03	0.660	0.509260	
clm_freq	2.661e+02	6.168e+01	4.314	1.63e-05	***
mvr_pts	2.290e+02	2.977e+01	7.692	1.66e-14	***
car_age	-3.416e+01	1.169e+01	-2.923	0.003475	**

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 4725 on 6514 degrees of freedom

Multiple R-squared: 0.03187, Adjusted R-squared: 0.02994

F-statistic: 16.5 on 13 and 6514 DF, p-value: < 2.2e-16

6.3 Model 3. Backward Stepwise Selection Models

For this model, we performed backward stepwise selection from the full model to identify the most important predictors.

The AIC for the stepwise logistic model is `r\ AIC(model3_logistic)`, and the R-squared for the stepwise linear model is `r\ summary(model3_linear)$r.squared`.

```
# Model 3: Backward Stepwise Logistic Model
full_logistic <- glm(
  formula = target_flag ~ .,
  data = insurance_train_split |> select(-target_amt),
  family = binomial
```

```
)

model3_logistic <- step(full_logistic, direction = "backward", trace = 0)

cat("\n=== Stepwise Logistic Model Summary ===\n")
```

```
=== Stepwise Logistic Model Summary ===
```

```
summary(model3_logistic)
```

Call:

```
glm(formula = target_flag ~ kidsdriv + age + yoj + travtime +
    tif + oldclaim + mvr_pts + parent1_yes + mstatus_z_no + education_bachelors +
    education_masters + education_phd + job_doctor + job_home_maker +
    job_manager + job_professional + job_student + car_use_private +
    car_type_panel_truck + car_type_pickup + car_type_sports_car +
    car_type_van + car_type_suv + revoked_yes + urbanicity_highly_rural +
    age_missing + home_val_missing + log_income + log_home_val +
    log_bluebook + log_oldclaim + age_bucket_older_driver + age_bucket_young_driver,
    family = binomial, data = select(insurance_train_split, -target_amt))
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)	
(Intercept)	6.822e+00	1.506e+00	4.529	5.92e-06	***
kidsdriv	4.458e-01	6.187e-02	7.205	5.82e-13	***
age	-8.248e-03	4.775e-03	-1.728	0.084068	.
yoj	1.730e-02	1.193e-02	1.450	0.146972	
travtime	1.651e-02	2.121e-03	7.785	6.95e-15	***
tif	-5.304e-02	8.281e-03	-6.405	1.50e-10	***
oldclaim	-2.843e-05	5.244e-06	-5.422	5.89e-08	***
mvr_pts	9.366e-02	1.582e-02	5.922	3.18e-09	***
parent1_yes	4.165e-01	1.129e-01	3.688	0.000226	***
mstatus_z_no	4.828e-01	9.308e-02	5.187	2.14e-07	***
education_bachelors	-4.629e-01	8.877e-02	-5.215	1.84e-07	***
education_masters	-6.072e-01	1.005e-01	-6.043	1.51e-09	***
education_phd	-8.093e-01	1.645e-01	-4.920	8.64e-07	***
job_doctor	-4.626e-01	2.937e-01	-1.575	0.115221	

```

job_home_maker      -4.048e-01  1.687e-01  -2.400  0.016413  *
job_manager         -7.864e-01  1.242e-01  -6.331  2.44e-10  ***
job_professional    -1.566e-01  1.105e-01  -1.418  0.156295
job_student         -4.131e-01  1.559e-01  -2.650  0.008059  **
car_use_private     -6.931e-01  8.427e-02  -8.225  < 2e-16  ***
car_type_panel_truck  6.116e-01  1.552e-01  3.941  8.11e-05  ***
car_type_pickup      5.774e-01  1.109e-01  5.207  1.91e-07  ***
car_type_sports_car  9.063e-01  1.226e-01  7.390  1.47e-13  ***
car_type_van         6.963e-01  1.362e-01  5.114  3.15e-07  ***
car_type_suv         7.223e-01  9.643e-02  7.491  6.85e-14  ***
revoked_yes         9.368e-01  1.044e-01  8.977  < 2e-16  ***
urbanicity_highly_rural -2.344e+00  1.273e-01 -18.412  < 2e-16  ***
age_missing         1.925e+00  1.220e+00  1.577  0.114814
home_val_missing     3.194e-01  8.744e-02  3.653  0.000259  ***
log_income          -8.588e-02  2.028e-02  -4.234  2.29e-05  ***
log_home_val        -3.525e-01  1.204e-01  -2.929  0.003402  **
log_bluebook        -3.376e-01  6.217e-02  -5.430  5.63e-08  ***
log_oldclaim        8.674e-02  1.106e-02  7.846  4.28e-15  ***
age_bucket_older_driver 8.352e-01  1.836e-01  4.550  5.36e-06  ***
age_bucket_young_driver 4.625e-01  3.260e-01  1.419  0.156018
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

(Dispersion parameter for binomial family taken to be 1)

```

Null deviance: 7531.0  on 6527  degrees of freedom
Residual deviance: 5759.5  on 6494  degrees of freedom
AIC: 5827.5

```

Number of Fisher Scoring iterations: 5

```

# Model 3: Backward Stepwise Linear Model
full_linear <- lm(
  formula = target_amt ~ .,
  data = insurance_train_split |> select(-target_flag)
)

model3_linear <- step(full_linear, direction = "backward", trace = 0)

```

```
cat("\n=== Stepwise Linear Model Summary ===\n")
```

```
=== Stepwise Linear Model Summary ===
```

```
summary(model3_linear)
```

Call:

```
lm(formula = target_amt ~ kidsdriv + income + tif + oldclaim +
    clm_freq + mvr_pts + car_age + parent1_yes + mstatus_z_no +
    female + job_doctor + job_manager + car_use_private + car_type_sports_car +
    car_type_van + car_type_suv + revoked_yes + urbanicity_highly_rural +
    log_bluebook + log_travtime + claim_rate + age_bucket_older_driver,
    data = select(insurance_train_split, -target_flag))
```

Residuals:

Min	1Q	Median	3Q	Max
-5661	-1702	-761	370	103447

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-2.607e+02	1.068e+03	-0.244	0.807097
kidsdriv	3.507e+02	1.158e+02	3.030	0.002456 **
income	-5.187e-03	1.519e-03	-3.415	0.000641 ***
tif	-4.178e+01	1.393e+01	-3.000	0.002713 **
oldclaim	-7.157e-02	2.564e-02	-2.791	0.005273 **
clm_freq	2.057e+02	6.904e+01	2.979	0.002900 **
mvr_pts	1.723e+02	2.962e+01	5.817	6.28e-09 ***
car_age	-3.336e+01	1.158e+01	-2.882	0.003970 **
parent1_yes	6.379e+02	2.014e+02	3.168	0.001543 **
mstatus_z_no	6.349e+02	1.370e+02	4.636	3.62e-06 ***
female	-2.883e+02	1.702e+02	-1.694	0.090321 .
job_doctor	-6.149e+02	3.673e+02	-1.674	0.094111 .
job_manager	-9.248e+02	1.857e+02	-4.980	6.51e-07 ***
car_use_private	-8.311e+02	1.310e+02	-6.343	2.41e-10 ***
car_type_sports_car	9.203e+02	2.393e+02	3.846	0.000121 ***
car_type_van	4.405e+02	2.122e+02	2.076	0.037935 *
car_type_suv	5.438e+02	1.862e+02	2.920	0.003511 **

```

revoked_yes          5.487e+02  1.982e+02   2.769 0.005640 **
urbanicity_highly_rural -1.586e+03  1.569e+02 -10.106 < 2e-16 ***
log_bluebook         1.617e+02  1.073e+02   1.507 0.131773
log_travtime         2.974e+02  1.013e+02   2.935 0.003347 **
claim_rate           1.609e-01  6.197e-02   2.596 0.009440 **
age_bucket_older_driver 4.797e+02  3.047e+02   1.575 0.115396
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

Residual standard error: 4638 on 6505 degrees of freedom

Multiple R-squared: 0.06857, Adjusted R-squared: 0.06542

F-statistic: 21.77 on 22 and 6505 DF, p-value: < 2.2e-16

Of these three models, the stepwise models have the lowest AIC for logistic and highest R-squared for linear. In the next section, we will evaluate the performance of these models to determine which ones are best for predicting the insurance outcomes.

7 Select Models

We evaluate both logistic and linear regression models using standard metrics from the textbook methods.

7.1 Evaluation Metrics Function

```

#' Evaluate Logistic Regression Models
#' @param model A glm logistic model
#' @param model_name Character name of the model
#' @param data Test dataframe
#' @param target_col Name of target column
#' @return Tibble with evaluation metrics
evaluate_logistic_model <- function(model, model_name, data, target_col = "target_flag") {
  probs <- predict(model, newdata = data, type = "response")
  preds <- factor(ifelse(probs > 0.5, 1, 0), levels = c(0, 1))
  actuals <- factor(data[[target_col]], levels = c(0, 1))

  cm <- confusionMatrix(preds, actuals, positive = "1")
  roc_obj <- roc(data[[target_col]], probs, quiet = TRUE)
  auc_val <- as.numeric(pROC::auc(roc_obj))

```



```

  tibble(
    Model = model_name,
    Deviance = model$deviance,
    Accuracy = cm$overall["Accuracy"],
    Error_Rate = 1 - cm$overall["Accuracy"],
    Precision = cm$byClass["Precision"],
    Sensitivity = cm$byClass["Sensitivity"],
    Specificity = cm$byClass["Specificity"],
    F1 = cm$byClass["F1"],
    AUC = auc_val,
    AIC = AIC(model),
    BIC = BIC(model)
  )
}

#' Evaluate Linear Regression Models
#' @param model An lm linear model
#' @param model_name Character name of the model
#' @param data Test dataframe
#' @param target_col Name of target column
#' @return Tibble with evaluation metrics
evaluate_linear_model <- function(model, model_name, data, target_col = "target_amt") {
  preds <- predict(model, newdata = data)
  actuals <- data[[target_col]]

  # Calculate metrics
  mse <- mean((actuals - preds)^2)
  rmse <- sqrt(mse)
  mae <- mean(abs(actuals - preds))
  r2 <- summary(model)$r.squared
  adj_r2 <- summary(model)$adj.r.squared

  tibble(
    Model = model_name,
    MSE = mse,
    RMSE = rmse,
    MAE = mae,
    R_Squared = r2,

```

```

      Adj_R_Squared = adj_r2,
      AIC = AIC(model),
      BIC = BIC(model)
    )
  }

# Evaluate all logistic models
cat("=== Evaluating Logistic Regression Models ===\n")

=== Evaluating Logistic Regression Models ===

logistic_models <- list(
  "Full Logistic Model" = model1_logistic,
  "Reduced Logistic Model" = model2_logistic,
  "Stepwise Logistic Model" = model3_logistic
)

logistic_comparison <- imap_dfr(logistic_models, ~ {
  evaluate_logistic_model(.x, .y, insurance_test_split)
}) |> arrange(AIC)

# Display logistic model comparison
logistic_comparison |>
  knitr::kable(digits = 4, caption = "Logistic Model Comparison")

```

Table 4: Logistic Model Comparison

Model	De- viance	Ac- cu- racy	Er- ror_Rate	Preci- sion	Sensi- tivity	Speci- ficity	F1	AUC	AIC	BIC
Stepwise Logistic Model	5759.498	0.7924	0.2076	0.6679	0.4282	0.9234	0.5219	0.8099	5827.498	6058.149
Full Logistic Model	5748.141	0.7961	0.2039	0.6800	0.4329	0.9267	0.5290	0.8104	5852.141	6204.901
Reduced Logistic Model	6803.144	0.7446	0.2554	0.5556	0.1736	0.9500	0.2646	0.6810	6831.144	6926.118

```
# Select best logistic model
best_logistic <- logistic_comparison |> slice(1)
cat("\nBest Logistic Model:", best_logistic$Model, "\n")
```

Best Logistic Model: Stepwise Logistic Model

```
# Evaluate all linear models
cat("\n=== Evaluating Linear Regression Models ===\n")
```

=== Evaluating Linear Regression Models ===

```
linear_models <- list(
  "Full Linear Model" = model1_linear,
  "Reduced Linear Model" = model2_linear,
  "Stepwise Linear Model" = model3_linear
)

linear_comparison <- imap_dfr(linear_models, ~ {
  evaluate_linear_model(.x, .y, insurance_test_split)
}) |> arrange(desc(R_Squared))

# Display linear model comparison
linear_comparison |>
  knitr::kable(digits = 4, caption = "Linear Model Comparison")
```

Table 5: Linear Model Comparison

Model	MSE	RMSE	MAE	R_Squared	Adj_R_Squared	AIC	BIC
Full Linear Model	17184817	4145.457	1928.511	0.0709	0.0636	128811.8	129171.4
Stepwise Linear Model	17094366	4134.533	1926.207	0.0686	0.0654	128770.2	128933.0
Reduced Linear Model	17873538	4227.711	2028.093	0.0319	0.0299	129004.5	129106.2

```
# Select best linear model
best_linear <- linear_comparison |> slice(1)
cat("\nBest Linear Model:", best_linear$Model, "\n")
```

Best Linear Model: Full Linear Model

For the logistic models, we use AIC as our primary criterion for model selection, with lower AIC indicating better model fit. The Stepwise Logistic Model achieved the lowest AIC of 5827.498296, with an accuracy of 0.792 and AUC of 0.81.

For the linear regression models, we use R-squared to select the best model, with higher R-squared indicating better model fit. The Full Linear Model achieved the highest R-squared of 0.071 with RMSE of 4145.46.

7.2 Selected Model Summaries

```
# Display summary of best logistic model
cat("=== Best Logistic Model Summary ===\n")
```

=== Best Logistic Model Summary ===

```
summary(logistic_models[[best_logistic$Model]])
```

Call:

```
glm(formula = target_flag ~ kidsdriv + age + yoj + travtime +
    tif + oldclaim + mvr_pts + parent1_yes + mstatus_z_no + education_bachelors +
    education_masters + education_phd + job_doctor + job_home_maker +
    job_manager + job_professional + job_student + car_use_private +
    car_type_panel_truck + car_type_pickup + car_type_sports_car +
    car_type_van + car_type_suv + revoked_yes + urbanicity_highly_rural +
    age_missing + home_val_missing + log_income + log_home_val +
    log_bluebook + log_oldclaim + age_bucket_older_driver + age_bucket_young_driver,
    family = binomial, data = select(insurance_train_split, -target_amt))
```

Coefficients:

Estimate Std. Error z value Pr(>|z|)

```

(Intercept)          6.822e+00  1.506e+00   4.529 5.92e-06 ***
kidsdriv             4.458e-01  6.187e-02   7.205 5.82e-13 ***
age                 -8.248e-03  4.775e-03  -1.728 0.084068 .
yoj                  1.730e-02  1.193e-02   1.450 0.146972
travtime             1.651e-02  2.121e-03   7.785 6.95e-15 ***
tif                 -5.304e-02  8.281e-03  -6.405 1.50e-10 ***
oldclaim            -2.843e-05  5.244e-06  -5.422 5.89e-08 ***
mvr_pts              9.366e-02  1.582e-02   5.922 3.18e-09 ***
parent1_yes          4.165e-01  1.129e-01   3.688 0.000226 ***
mstatus_z_no         4.828e-01  9.308e-02   5.187 2.14e-07 ***
education_bachelors -4.629e-01  8.877e-02  -5.215 1.84e-07 ***
education_masters    -6.072e-01  1.005e-01  -6.043 1.51e-09 ***
education_phd        -8.093e-01  1.645e-01  -4.920 8.64e-07 ***
job_doctor           -4.626e-01  2.937e-01  -1.575 0.115221
job_home_maker       -4.048e-01  1.687e-01  -2.400 0.016413 *
job_manager          -7.864e-01  1.242e-01  -6.331 2.44e-10 ***
job_professional     -1.566e-01  1.105e-01  -1.418 0.156295
job_student          -4.131e-01  1.559e-01  -2.650 0.008059 **
car_use_private      -6.931e-01  8.427e-02  -8.225 < 2e-16 ***
car_type_panel_truck  6.116e-01  1.552e-01   3.941 8.11e-05 ***
car_type_pickup       5.774e-01  1.109e-01   5.207 1.91e-07 ***
car_type_sports_car   9.063e-01  1.226e-01   7.390 1.47e-13 ***
car_type_van          6.963e-01  1.362e-01   5.114 3.15e-07 ***
car_type_suv          7.223e-01  9.643e-02   7.491 6.85e-14 ***
revoked_yes          9.368e-01  1.044e-01   8.977 < 2e-16 ***
urbanicity_highly_rural -2.344e+00  1.273e-01 -18.412 < 2e-16 ***
age_missing           1.925e+00  1.220e+00   1.577 0.114814
home_val_missing      3.194e-01  8.744e-02   3.653 0.000259 ***
log_income           -8.588e-02  2.028e-02  -4.234 2.29e-05 ***
log_home_val         -3.525e-01  1.204e-01  -2.929 0.003402 **
log_bluebook         -3.376e-01  6.217e-02  -5.430 5.63e-08 ***
log_oldclaim          8.674e-02  1.106e-02   7.846 4.28e-15 ***
age_bucket_older_driver 8.352e-01  1.836e-01   4.550 5.36e-06 ***
age_bucket_young_driver 4.625e-01  3.260e-01   1.419 0.156018
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 7531.0 on 6527 degrees of freedom
 Residual deviance: 5759.5 on 6494 degrees of freedom
 AIC: 5827.5

Number of Fisher Scoring iterations: 5

```
# Display summary of best linear model
cat("\n=== Best Linear Model Summary ===\n")
```

=== Best Linear Model Summary ===

```
summary(linear_models[[best_linear$Model]])
```

Call:

```
lm(formula = target_amt ~ ., data = select(insurance_train_split,
      -target_flag))
```

Residuals:

Min	1Q	Median	3Q	Max
-5476	-1694	-760	387	103433

Coefficients: (1 not defined because of singularities)

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	9.078e+02	7.454e+03	0.122	0.90307
kidsdriv	3.108e+02	1.287e+02	2.414	0.01579 *
age	-1.282e-01	9.262e+00	-0.014	0.98896
homekids	3.280e+01	7.557e+01	0.434	0.66433
yoj	-1.116e+00	2.189e+01	-0.051	0.95936
income	-2.254e-03	2.672e-03	-0.843	0.39899
home_val	-6.342e-04	3.161e-03	-0.201	0.84098
travtime	4.353e+00	1.018e+01	0.427	0.66908
bluebook	9.426e-04	2.091e-02	0.045	0.96405
tif	-4.269e+01	1.400e+01	-3.049	0.00230 **
oldclaim	-6.465e-02	2.699e-02	-2.396	0.01662 *
clm_freq	1.338e+02	1.288e+02	1.038	0.29910
mvr_pts	1.633e+02	3.062e+01	5.333	9.97e-08 ***
car_age	-1.753e+01	2.066e+01	-0.848	0.39626

parent1_yes	5.603e+02	2.313e+02	2.422	0.01545	*
mstatus_z_no	5.819e+02	1.679e+02	3.467	0.00053	***
female	-3.282e+02	2.135e+02	-1.537	0.12437	
education_bachelors	-2.888e+02	2.391e+02	-1.208	0.22723	
education_masters	3.296e+00	3.498e+02	0.009	0.99248	
education_phd	-2.056e+02	4.151e+02	-0.495	0.62046	
education_high_school	-1.813e+02	1.997e+02	-0.908	0.36405	
job_doctor	-9.151e+02	5.033e+02	-1.818	0.06908	.
job_home_maker	-4.712e+02	3.171e+02	-1.486	0.13727	
job_lawyer	-4.518e+02	3.503e+02	-1.290	0.19716	
job_manager	-1.158e+03	2.726e+02	-4.247	2.20e-05	***
job_professional	-1.423e+02	2.485e+02	-0.573	0.56696	
job_student	-4.002e+02	3.120e+02	-1.283	0.19962	
job_blue-collar	-8.843e+01	2.221e+02	-0.398	0.69060	
job_na	-7.612e+02	3.922e+02	-1.941	0.05235	.
car_use_private	-8.037e+02	1.889e+02	-4.255	2.12e-05	***
car_type_panel_truck	2.656e+02	3.251e+02	0.817	0.41401	
car_type_pickup	2.608e+02	1.948e+02	1.339	0.18074	
car_type_sports_car	1.009e+03	2.553e+02	3.952	7.85e-05	***
car_type_van	5.800e+02	2.449e+02	2.369	0.01787	*
car_type_suv	6.435e+02	2.084e+02	3.088	0.00202	**
red_car_yes	-8.997e+01	1.724e+02	-0.522	0.60185	
revoked_yes	5.471e+02	2.003e+02	2.731	0.00633	**
urbanicity_highly_rural	-1.623e+03	1.617e+02	-10.038	< 2e-16	***
age_missing	1.162e+03	2.095e+03	0.555	0.57892	
car_age_missing	7.637e+01	2.433e+02	0.314	0.75362	
home_val_missing	1.516e+02	1.741e+02	0.871	0.38383	
income_missing	1.257e+02	2.608e+02	0.482	0.62983	
voj_missing	3.243e+02	2.523e+02	1.285	0.19869	
log_income	-3.847e+01	4.027e+01	-0.955	0.33946	
log_home_val	-1.993e+01	6.518e+02	-0.031	0.97561	
log_bluebook	1.353e+02	2.453e+02	0.551	0.58139	
log_oldclaim	2.420e+01	3.927e+01	0.616	0.53773	
log_travtime	1.811e+02	2.828e+02	0.640	0.52189	
claim_rate	1.335e-01	7.295e-02	1.830	0.06727	.
car_value_age_ratio	1.344e-02	3.263e-02	0.412	0.68031	
risk_points_claims	NA	NA	NA	NA	
age_bucket_older_driver	5.248e+02	3.444e+02	1.524	0.12758	
age_bucket_young_driver	3.244e+02	6.563e+02	0.494	0.62111	

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 4643 on 6476 degrees of freedom

Multiple R-squared: 0.0709, Adjusted R-squared: 0.06359

F-statistic: 9.69 on 51 and 6476 DF, p-value: < 2.2e-16

To make inferences from our top performing models, we can look at the significance of the predictors.

For the logistic regression model (target_flag), the model identifies key factors that significantly predict whether a person will be in a car crash. The most important predictors include variables related to driving history, vehicle characteristics, and demographic factors.

For the linear regression model (target_amt), significant predictors help explain the variation in the cost of claims when crashes occur.

8 Results

Generate predictions on the evaluation dataset using the selected models:

```
# Get the best model names
best_logistic_name <- best_logistic$Model
best_linear_name <- best_linear$Model

# Generate predictions for target_flag (probability of crash)
eval_probs <- predict(
  logistic_models[[best_logistic_name]],
  newdata = insurance_eval_prepared,
  type = "response"
)

# Generate predictions for target_amt (cost if crash)
eval_amounts <- predict(
  linear_models[[best_linear_name]],
  newdata = insurance_eval_prepared
)

# Create results dataframe
insurance_eval_results <- insurance_eval_prepared |>
  mutate(
```



```

    PREDICTED_target_flag = ifelse(eval_probs > 0.5, 1, 0),
    PREDICTED_target_amt = ifelse(PREDICTED_target_flag == 1,
                                  round(eval_amounts, 2),
                                  0)

  )

# Write predictions to CSV
write_csv(insurance_eval_results, "insurance_evaluation_predictions.csv")

# Display sample predictions
cat("Sample predictions for evaluation data:\n")

```

Sample predictions for evaluation data:

```

insurance_eval_results |>
  select(PREDICTED_target_flag, PREDICTED_target_amt) |>
  head(10) |>
  knitr::kable(caption = "Sample Insurance Predictions")

```

Table 6: Sample Insurance Predictions

PREDICTED_target_flag	PREDICTED_target_amt
1	1392.17
1	2220.30
1	1347.85
1	2238.33
1	1129.25
0	0.00
1	2845.22
0	0.00
0	0.00
1	2029.34

We generate predictions for both outcomes:

1. **target_flag**: Whether the person will be in a car crash (binary: 0 or 1)
2. **target_amt**: The cost if a crash occurs (continuous dollar amount)

The predictions are written to `insurance_evaluation_predictions.csv` for submission.

9 Conclusion

In conclusion, our `best_model_info$Model` model, which was selected based on deviance and other performance metrics, demonstrates strong predictive capabilities for classifying neighborhoods as high-crime or low-crime based on a variety of environmental and structural factors. The significant predictors identified in the model provide valuable insights into the underlying drivers of crime in urban areas, highlighting the importance of air pollution, school crowding, aging infrastructure, and proximity to employment centers. These findings can inform policymakers and community leaders in their efforts to target interventions and allocate resources effectively to reduce crime rates. Future work could explore additional modeling techniques, such as regularization or ensemble methods, to further enhance predictive performance and interpretability.